WEEK 9: ORGAN SYSTEM

DIGESTIVE SYSTEM

FUNCTIONS

1.) Ingestion of food - food and water enter the body through the mouth
2.) Digestion of food - food is broken down from complex particles to smaller molecules via mechanical and chemical digestion
3.) Absorption of nutrients - absorbed mainly in the small intestine
4.) Elimination of wastes - undigested material, such as fiber from food, plus waste products excreted into the GI tract are eliminated in the feces

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GI SYSTEM

- Consists of the alimentary canal (digestive tract) + accessory organs
- **Alimentary:** a continuous, hollow tube, about 7-9 meters long, extending from the mouth to the anus
- **Accessory organs**: structures that secrete enzymes, bile, and provide mechanical assistance in digestion (not a pathway)
    - ***Liver*** - bile production, hypercholestorelemia
    - ***Gallbladder*** - stores and releases bile

- **Ligament of Treitz** - separates the upper & lower GI tract

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ALIMENTARY CANAL ANATOMY

Mouth "*oral cavity*"

- First part of the digestive tract; entry point of food
- Mechanical (start) and chemical digestion, speech, and defense for immune function

STRUCTURAL COMPOSITION:

- **Lips**
    - Forms the anterior boundary of the mouth
    - Highly vascular = reddish appearance
    - Muscular structures, formed mostly by the ***orbicularis oris*** (functions as a sphincter, mostly for closing, making the pouting facial expressions)
    - Bell's palsy (CN7)
        - Ipsilateral manifestation
        - Lips and cheeks
    - Ulcer
        - Aphtous
- **Cheeks**
    - Forms the lateral walls of the oral cavity
    - ***Buccinator*** - muscle located within the cheeks and keeps the food between the teeth during chewing and prevent food from spilling out laterally
        - For mastication
- **Palate**
    - Forms the roof of the mouth and separates the oral cavity from the nasal cavity
    - Consists of two parts:
        - **Hard palate** - anterior part; *contains bone* -> palatine process of maxilla and horizontal plates of the palatine bone (rigid surface)
        - **Soft palate** - posterior part; *contains muscle* -> closes the nasopharynx to prevent the food from entering the nasal cavity
        - **Uvula** - *posterior extension of the soft palate*
- **Tongue**
    - Muscular organ with skeletal fibers for strength, flexibility, and precise movement
    - Covered with *papillae* -> small projections containing taste buds (except filiform = more on friction)

WEEK 9: ORGAN SYSTEM

- **Anterior ⅔ (CN 7)** - covered by papillae, some of which contain taste buds
- **Posterior ⅓ (CN 9)** - devoid of papillae and has only a few scattered taste buds
- **<u>Functions:</u>**
  - Mastication by moving food around
  - Deglutition helps food push posteriorly into the pharynx (from food to bolus)
  - Articulation and speech
  - Sensory: taste and tactile

Sweet	Anterior
Bitter	Posterior
Sour	Lateral
Salty	Anterolateral
Umami	All over
Spicy	All over

- **Teeth**
  - ***32 teeth***: <u>embedded alveolar processes of mandible and maxillae</u>
    - **8 incisors** - *chiseled*; cutting
      - Front
    - **4 canines** - *pointed*; tearing and piercing
      - Anterolateral (cuspid)
    - **8 premolars** - *crowns*; crushing and grinding
      - Bicuspids
    - **12 molars (including 4 wisdom teeth)** - *large, flat surfaces*; grinding and chewing
  - Mainly for mechanical digestion

FUNCTIONS OF MOUTH

1.) Mechanical breakdown
- <u>Teeth:</u> cut, tear, and grind food
- <u>Tongue and cheeks:</u> manipulate food, forming into bolus suitable for swallowing

2.) Mixing with saliva
- Contains <u>mucus</u> (lubrication) and <u>salivary amylase</u> (begins the chemical digestion of carbs, *mostly starch*)
- Moistens bolus

3.) Initiation of deglutition
- Tongue pushes bolus posteriorly towards pharynx
- Partly voluntary and reflexive

4.) Additional roles
- <u>Speech</u> - shaping sounds with lips, teeth, and tongue
- <u>Taste</u> - taste buds
- <u>Defense</u> - tonsils

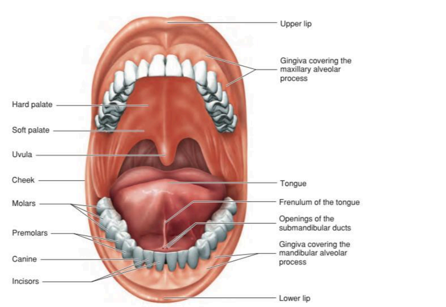


Figure 16.4 AP|R Oral Cavity

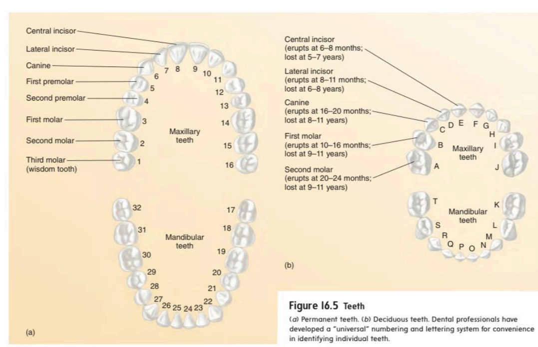


Figure 16.5 Teeth
(*a*) Permanent teeth. (*b*) Deciduous teeth. Dental professionals have developed a "universal" numbering and lettering system for convenience in identifying individual teeth.

Pharynx "throat"

- Funnel-shaped tube: 12-14 cm long

WEEK 9: ORGAN SYSTEM

- From base of the skull (C1), which continues inferiorly as esophagus

- ***Pharyngeal constrictor muscles*** - skeletal walls with mucous membrane
- Common pathway for air, food, and fluids; consist of 3 parts:
  - ***Nasopharynx***
    - Uppermost, behind nasal cavity, above soft palate
    - Function: strictly part of respiratory system
    - Features:
      - *Pharyngeal tonsil (adenoids)* - immune defense
      - *Eustachian (auditory) tubes* - equalize air pressure between nasopharynx and middle ear
  - ***Oropharynx***
    - Middle, behind oral cavity, extends from the soft palate to the level of hyoid bone
    - Function: common passageway for food and air
    - Features:
      - *Palatine tonsil (tonsilectomy)* - usually inflamed
      - *Lingual tonsil* - base of the tongue
      - *Fauces* - opening between oral cavity and pharynx
  - ***Laryngopharynx***
    - Lowermost, extending from the hyoid bone to the cricoid cartilage
    - Function: passageway for both food and air
    - Features:
      - Directly behind larynx
      - *Epiglottis* - flap of elastic cartilage which closes larynx during deglutition

FUNCTIONAL ROLE OF PHARYNX IN DIGESTION

- Pharynx propel bolus from mouth to esophagus
  - Achieved by swallow reflex: **CN 9 & 10 -> medulla**

1.) Voluntary phase
- Tongue pushes bolus to the oropharynx

2.) Pharyngeal phase
- *Involuntary reflex*; soft palate rises to block nasopharynx, epiglottis folds down to cover larynx, and the pharyngeal constrictor muscles contract sequentially to push food downward

3.) Esophageal phase
- Bolus enters the esophagus; *peristalsis*

CLINICAL RELEVANCE

1.) Aspiration
- Swallow reflex fails, leading the bolus to enter airway

2.) Tonsillitis (infection of palatine tonsils)
- Painful swallowing and obstruction

3.) Obstructive sleep apnea
- Collapse of pharyngeal airway during sleep

Esophagus

- Muscular tube, lacks serosa: 25 cm or 10 inches long
- From C6 to T10:
  - From cricoid cartilage (C6), down through the thoracic cavity, passing through the diaphragm at the esophageal hiatus (T10), ending at the cardiac orifice of the stomach, guarded by LES

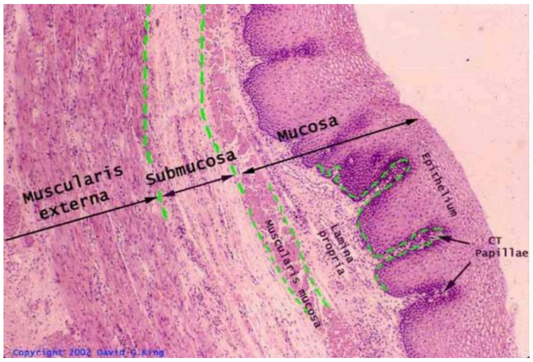
SPHINCTERS OF THE ESOPHAGUS:

- ***Upper esophageal sphincter (UES)***
  - Location: junction of pharynx and esophagus
  - Formed by *cricopharyngeus muscle*
  - Prevents air from entering esophagus and regulates passage of bolus during deglutition
- ***Lower esophageal sphincter (LES)***
  - Location: GE junction (T11)

WEEK 9: ORGAN SYSTEM

- Prevents reflux of acidic gastric contents into the esophagus
- <u>Dysfunction:</u> GERD

HISTOLOGY OF THE ESOPHAGUS



1.) Mucosa

- Contains non-keratinized stratified squamous epithelium
    - Protection against abrasion

- Consist of <u>connective tissue (lamina propria)</u> and a <u>thin layer of smooth muscle (muscularis mucosae)</u>

2.) Submucosa

- Contains esophageal glands (secret mucus for bolus lubrication)
- For elasticity, allowing expansion during swallowing

3.) Muscularis externa

- Contains inner circular and outer longitudinal layers of muscle
    - <u>Upper ⅓</u> - *voluntary*; skeletal - initiation of swallowing
    - <u>Middle ⅓</u> - *mixed*
    - <u>Lower ⅓</u> - *involuntary*; smooth muscle - controlled by ANS (CN 10)
        - Adventitia: upper and middle
        - Serosa: lower

- This transition allows swallowing to start under voluntary control and the continue reflexively -> <u>peristaltic contraction</u>

4.) Adventitia/Serosa

- Esophagus is mostly surrounded by adventitia (connective tissue attaching it to surrounding structures)
- Only a short portion within abdominal cavity is covered by serosa (visceral peritoneum)

FUNCTIONS OF ESOPHAGUS

1.) Only function: Transporting bolus from pharynx to stomach

- Through peristalsis (wave-like, coordinated contractions of the muscular wall that push bolus downward)
- Gravity also assists when upright, but it's mostly peristalsis

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CLINICAL RELEVANCE

1.) GERD

- Failure of LES to close properly, which allows acid reflux into the esophagus, causing heartburn

2.) Barrett's esophagus

- Chronic reflux; precancerous condition
- Non-keratinized to simple squamous epithelium (mucus) -> counteracts with acid

3.) Achalasia

- Failure of LES to relax properly, leading to difficulty in swallowing (dysphagia), regurgitation, and esophageal dilation

4.) Esophageal varices

- Dilated veins in the lower esophagus due to portal hypertension, risk of massive hemorrhage

5.) Hiatal hernia

- Stomach herniates upward through the esophageal hiatus, potentially contributing to reflux

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Stomach

WEEK 9: ORGAN SYSTEM

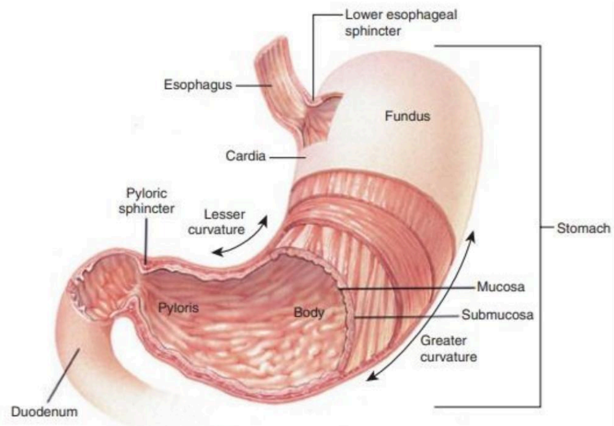


FIGURE The stomach.

- Lies at epigastric and left hypochondriac region
  - Inferior to esophagus and superior to duodenum
  - ***When empty:*** 50 mL, but can expand up to 1 to 1.5 L
- Produces **chyme (semi-liquid balance)**
  - Muscular contraction + gastric secretion

DIVISIONS OF THE STOMACH:

1.) Cardia

- Area surrounding the cardiac orifice, where food enters from the esophagus
- Contains cardiac sphincter/LES (functional, not anatomical) to prevent reflux

2.) Fundus

- Dome-shaped portion superior to the cardia
- Usually filled with gas in radiographs
  - Accumulates swallowed air

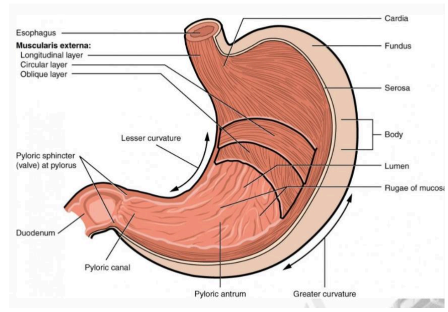
3.) Body

- Largest and central region, where most mixing and secretion occur

4.) Pylorus

- Funnel-shaped distal portion, divided into:
  - Pyloric antrum (wider part)
  - Pyloric canal (narrower part leading to the small intestine)
  - Pyloric sphincter (end part; a strong circular muscle emptying into the duodenum)

STRUCTURAL FEATURES



1.) Rugae

- Internal folds of the mucosa that allow the stomach to expand significantly after a meal

2.) Gastric pits

- Invaginations in the mucosa leading to gastric glands, which secrete gastric juice

3.) Muscularis externa

- Unlike the rest of the GI, stomach has 3 muscle layers which enhances churning and mixing of food:
  - Inner oblique, middle circular, and outer longitudinal

HISTOLOGY OF THE GASTRIC MUCOSA

1.) Parietal cells

- Secrete HCl - lowers pH, denatures proteins, and activates *pepsinogen*
- Secrete intrinsic factor (vitamin B12 absorption)

2.) Mucous cells

- Secrete mucus that protects the stomach lining from self-digestion by acid
- Produce cyclooxygenase + prostaglandin for stomach lining protection

3.) Chief cells

- Secrete pepsinogen - enzyme that is converted to pepsin in the presence of HCl
  - Pepsin digests proteins

4.) Enteroendocrine (G cells)

WEEK 9: ORGAN SYSTEM

- Secrete gastrin - hormone that stimulates gastric acid secretion and motility

FUNCTIONS OF THE STOMACH

1.) Temporary storage

- Holds ingested food and regulates its release into the small intestine

2.) Mechanical digestion

- Muscular contraction + gastric secretion = chyme (semi-liquid balance)

3.) Chemical digestion

- Gastric juice breaks down food chemically
  - HCl - denatures protein, kills pathogen
  - Pepsin - begins protein digestion
  - Lipase (minor role) - contributes slightly to fat digestion

4.) Secretion of intrinsic factor

- Required for vitamin B12 absorption in the ileum

5.) Defense

- Acidic environment = barrier to ingested microorganism

CLINICAL RELEVANCE

1.) Peptic ulcer

- Erosion of the stomach lining due to imbalance between acid and protective mucus
- Often associated with Helicobacter pylori infection or NSAID use

2.) Gastritis

- Inflammation of the gastric mucosa

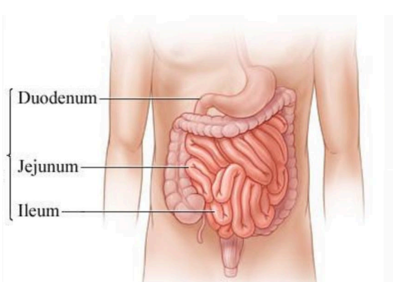
3.) GERD

- LES dysfunction = acid reflux

4.) Pernicious anemia

- Lack of intrinsic factor = vitamin B12 deficiency

Small Intestine



- Longest portion of the GI tract: 6 meters (20 feet)
- Primary site for digestion and absorption

- From pyloric sphincter of the stomach to the ileocecal valve, where it joins the large intestine
- Where segmentation happens, and completion of chemical digestion
  - Unabsorbed moves to large intestine

SUBDIVISIONS OF THE SMALL INTESTINE

1.) Duodenum (~25 cm / 10 inches)

- First and shortest portion, C-shaped, wrapping around the head of the pancreas
- Receives chyme from the stomach and secretion from the accessory organs:
  - **Bile (from liver and gallbladder)** - emulsifies fats
  - **Pancreatic juice (enzymes + bicarbonate)** - digests carbohydrates, proteins, and lipids; neutralizes acidic chyme
- Contains **Brunner's glands** (submucosa) - secrete alkaline mucus to protect mucosa from acidic gastric contents

2.) Jejunum (~2.5 m / 8 feet)

- Middle portion, located in the upper left abdomen
- Thick wall, highly vascular, and many plicae circulares (circular folds)
- Main site of nutrient absorption

3.) Ilieum (~3.5 m / 12 feet)

- Longest portion, located mainly in the lower right abdomen
- Absorbs vitamin B12 (with intrinsic factor), bile salts, and any remaining nutrients
- Contains **Peyer's patches** (aggregated lymphoid nodule) - part of the gut-associated lymphoid tissue (GALT), important for immune defense
- Contains **ileocecal valve** - connection between ileum and large intestine

SURFACE MODIFICATIONS OF ABSORPTION

1.) Plicae circulares (circular folds)

- Large, permanent transverse folds of mucosa and submucosa

WEEK 9: ORGAN SYSTEM

- Slow down movement of chyme, increasing contact time for absorption

2.) Villi
- Finger-like projections of mucosa (~1 mm tall)
- Each villus contains:
  - **Capillary network** - absorbs amino acids, sugars, and water-soluble vitamins
  - **Lacteal (lymphatic vessel)** - absorbs dietary lipids and fat-soluble vitamins (A, D, E, K)

3.) Microvilli ("brush border")
- Microscopic projections on the apical surface of absorptive epithelial cells
- Form the brush border, containing digestive enzymes (e.g., lactase, maltase, peptidase) for final breakdown of nutrients

FUNCTIONS OF SMALL INTESTINE
1.) Digestion:
- Carbohydrates - broken down into monosaccharides
- Proteins - reduced amino acids
- Fats - emulsified by bile, digested into fatty acids and glycerol
2.) Absorption
- About 90% of water and electrolyte absorption
3.) Immune defense
- Peyer's patches help monitor intestinal bacteria and prevent overgrowth of pathogens

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CLINICAL RELEVANCE
1.) Celiac disease
- Autoimmune reaction to gluten -> villous atrophy -> malabsorption
2.) Lactose intolerance
- Deficiency of lactase enzyme in brush border, causing bloating, diarrhea after dairy intake
3.) Crohn's disease
- Chronic inflammatory condition that often affects the ileum
4.) Small bowel obstruction
- Causes abdominal pain, vomiting, and constipation due to adhesions, hernia, or tumor

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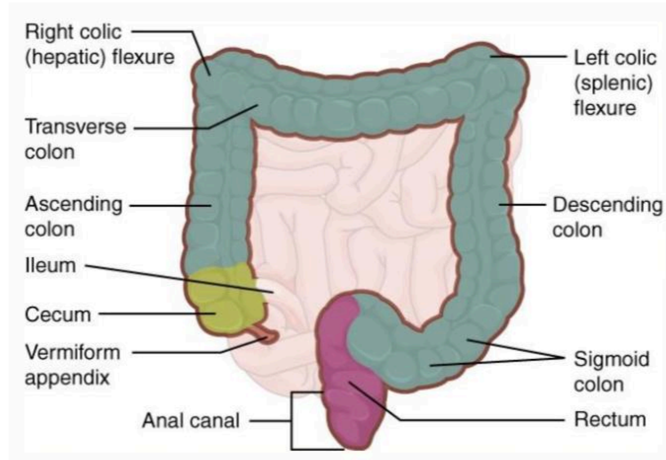
Large Intestine



- Extends from the ileocecal junction to the anus: 1.5 meters (5 feet)
- Functions more in absorption of water and electrolytes rather than digestion

DIVISIONS OF THE LARGE INTESTINE
1.) Cecum
- Pouch-like structure at the beginning
- Receives chyme through ileocecal valve (prevents backflow)
- **Vermiform appendix** - attached to the cecum which is a narrow, worm-shaped tube rich in lymphoid tissue, thought to play a role in immunity

2.) Colon
- Longest part, subdivided into 4 regions:
  - **Ascending colon** - travels upward along the right side of the abdomen
  - **Transverse colon** - runs across the abdominal cavity
  - **Descending colon** - travels downward on the left side
  - **Sigmoid colon** - S-shaped region that leads into the rectum

3.) Rectum
- Straight muscular tube: 12 cm long
- Temporary storage site for feces before defecation
- Contains rectal valves that separate feces from gas passage

WEEK 9: ORGAN SYSTEM

4,) Anal canal
- Terminal portion: ~3-4 cm long
- Surrounded by 2 sphincters:
  - **Internal anal sphincter**
    - Involuntary; smooth muscle
    - Maintains resting tone to prevent leakage
  - **External anal sphincter**
    - Voluntary, skeletal muscle
    - Under conscious control for defecation

DISTINGUISHING FEATURES OF LARGE INTESTINE

1.) Taeniae coli - "bulge"
- 3 distinct longitudinal bands of smooth muscle running along the colon

2.) Haustra - "inside the bulge"
- Sac-like pouches caused by contraction of taeniae coli, giving the colon its segmented appearance

3.) Epiploic appendages
- Small, fat-filled pouches of peritoneum hanging from the colon for cushioning or protection

FUNCTIONS OF LARGE INTESTINE

1.) Absorption of water & electrolytes
- Compacts liquid chyme into solid feces

2.) Vitamin synthesis
- Intestinal microbiota (gut flora) produce:
  - Vitamin K - clotting factor synthesis
  - Vitamin B complex - B7, B12, folate

3.) Formation & storage of feces
- Collects undigested material, mucus, dead cells, and bacteria into fecal matter

4.) Immune & protective function
- Appendix and gut flora (E. coli) contribute to immune defense

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CLINICAL RELEVANCE

1.) Appendicitis
- Inflammation of appendix; requires ectomy

2.) Diverticulosis / diverticulitis
- Outpouching (diverticula) in the colon wall, which can become inflamed

3.) Colorectal cancer
- Often develops from polyps; common in sigmoid colon and rectum

4.) Irritable bowel syndrome (IBS)
- Functional disorder with abdominal pain, bloating, and altered bowel habits

5.) Constipation / Diarrhea
- Dysfunction in water absorption

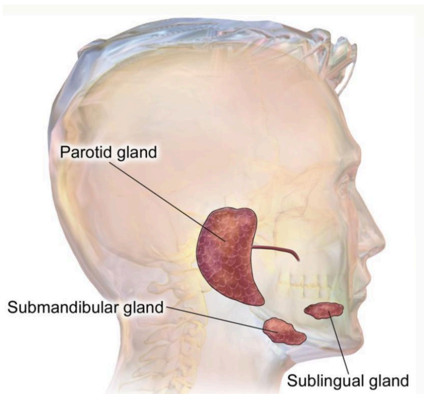
WEEK 9: ORGAN SYSTEM

ACCESSORY DIGESTIVE ORGANS

Salivary Glands

- Exocrine glands that produce saliva -> important for the initiation of digestion, lubrication of food, and maintenance of oral health
- **Average adult:** 1-1.5 L of saliva per day
    - Controlled by ANS parasympathetic

MAJOR SALIVARY GLANDS



Salivary Glands

1.) Parotid glands - "largest"
- **Location:** anterior and inferior to the ears, overlying masseter muscle
- **Secretion:** serous
- **Duct:** Stensen's duct (opens opposite the upper molar)
- **Clinical note:** Site of inflammation in mumps (viral infection)

2.) Submandibular glands
- **Location:** medial aspect of the mandible, beneath the jaw
- **Secretion:** both serous and mucus (mostly serous)
- **Duct:** Wharton's duct (opens on either side of the lingual frenulum on the mouth's floor) -> site for fast absorption of drugs (e.g., hypertensive drug)
- Contributes to 70% of total saliva production

3.) Sublingual glands - "smallest"
- **Location:** beneath the tongue
- **Secretion:** mucus
- **Ducts:** Rivinus' ducts (opens into the floor of the mouth)

4.) Minor salivary glands
- **Location:** scattered throughout oral mucosa, lips, cheeks, palate, and tongue
- Contribute to continuous low-level saliva secretion, especially important during rest or sleep

FUNCTIONS OF SALIVA

1.) Digestion
- Contains salivary amylase (ptyalin)
    - Begins the chemical digestion of starch into maltose

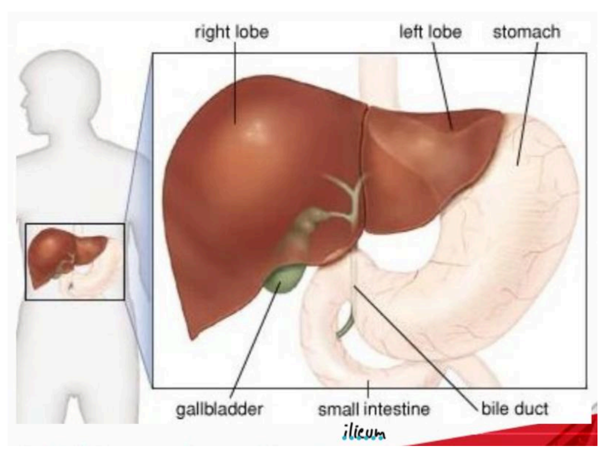
2.) Lubrication
- Through mucus for facilitation swallowing

3.) Taste
- Dissolves food molecules, allowing taste buds to detect flavors

4.) Protection
- Contains lysozyme (antibacterial enzyme)
- Rich in IgA antibodies for immune defense
- Helps maintain oral pH and wash away debris

5.) Speech & oral comfort
- Keeps mucous membranes moist, preventing dryness (xerotosmia)

Liver

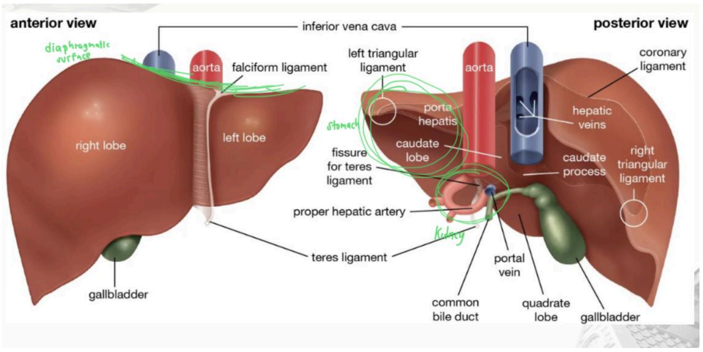


- Located in the right upper quadrant, beneath the diaphragm, partially protected by the rib cage (11th & 12th): 1.5 kg
- **Largest gland and internal organ of the human body**

WEEK 9: ORGAN SYSTEM

- Vital metabolic hub
    - Digestion, detoxification, storage, and synthesis of essential proteins

GROSS ANATOMY



1.) Location
- Right hypochondriac and epigastric region

2.) Lobes
- 4 anatomical lobes:
    - Right lobe (largest), left lobe, caudate lobe (posterior; stomach), quadrate lobe (anteroinferior)

3.) Surface
- Diaphragmatic surface - smooth, convex, in contact with diaphragm
- Visceral surface - faces abdominal organs, includes impressions from stomach, gallbladder, duodenum, right kidney, and colon **(SGDRC)**

4.) Ligaments
- Falciform ligament - attaches liver to the anterior abdominal wall
- Round ligament (ligamentum teres) - remnant of fetal umbilical vein

5.) Blood supply
- Dual supply:
    - Hepatic artery: ~25% oxygen-rich blood
    - Portal vein: ~75% nutrient rich blood from GI tract
- Blood drains via hepatic veins into the inferior vena cava

6.) Portal Triad - "nutrient supplement"
- Each corner of a lobule contains:
    - Portal vein - nutrients
    - Hepatic artery - oxygenation
    - Bile duct - bile made by hepatocytes, which will be stored in gallbladder

FUNCTIONS OF THE LIVER

1.) Digestive function
- Produces bile: 600-1000 mL per day
    - Contains bile salts, bilirubin, cholesterol, and electrolytes **(BBCE)**

2.) Metabolism
- Carbohydrates: *maintains blood glucose* via glycogenesis (storage), glycogenolysis (release), and gluconeogenesis (formation from non-carbohydrates)
- Lipids: converts excess carbohydrates/proteins into fatty acids and triglycerides; produces cholesterol and lipoproteins
- Proteins: Deaminates amino acids, converts ammonia into urea for excretion

3.) Storage
- Stores glycogen, fat-soluble vitamins (A, D, E, K), vitamin B12, and iron (ferritin)

4.) Detoxification
- Metabolizes drugs, alcohol, hormones
- Remove toxin and old RBC

5.) Synthesis of plasma proteins
- Produces albumin (oncotic pressure), clotting factors (fibrinogen, prothrombin), transport proteins, and complement proteins

6.) Immune function
- **Kupffer cells** - act as macrophage for the portal blood

CLINICAL RELEVANCE

1.) Jaundice
- Yellow discoloration due to buildup of bilirubin (liver disease, bile obstruction, or hemolysis)
    - Biliverdin - green discoloration

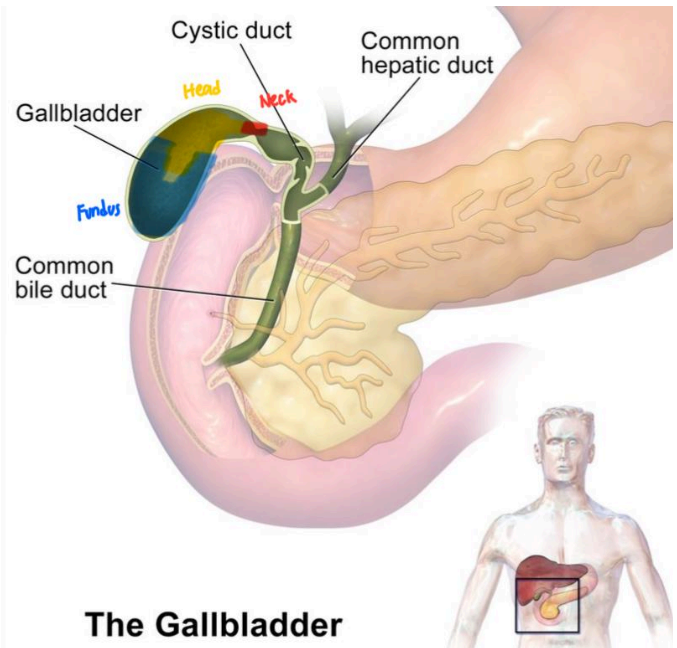
2.) Cirrhosis
- Chronic liver damage leading to fibrosis (scar tissue formation), nodular regeneration, and portal hypertension (common in alcoholism, hepatitis)

3.) Hepatitis
- Inflammation of liver due to viral infections,

WEEK 9: ORGAN SYSTEM

toxins, or autoimmune causes
4.) Fatty liver disease
- Accumulation of fat within hepatocytes, often linked with obesity. diabetes, or alcohol
5.) Liver failure
- Leads to impaired detoxification, bleeding disorders, hypoglycemia, and encephalopathy (disrupted brain function)

Gallbladder



- Small, pear-shaped sac: 7-10 cm long with ~30-40 mL capacity
    - Lies on the inferior surface of the liver, nestle in a fossa between right and quadrate lobes
- Stores, concentrates, and releases bile from liver
    - **Bile** - important for fat digestion or absorption

PARTS OF THE GALLBLADDER
1.) Fundus - rounded end projecting slightly beyond the liver's edge
2.) Body - main portion lying against the visceral surface of the liver
3.) Neck - narrow portion leading to cystic duct

DUCT SYSTEM
- **Biliary tree** - connection of gallbladder to liver and pancreas
    - **Cystic duct (gallbladder)** joins with **common hepatic duct (liver)**, forming **common bile duct (CBD)**
    - CBD travels downward and joins **pancreatic duct**
    - Together, they **open into the duodenum of the hepatopancreatic ampulla (ampulla of Vater)**, regulated by the **sphincter of Oddi**

FUNCTIONS OF GALLBLADDER
1.) Bile storage
- Produced bile in a day: ~600-1000 mL/day
- Gallbladder stores the excess until needed
2.) Bile concentration
- Absorbs water and electrolytes from stored bile, concentrates it 5-10 times
    - Makes bile more effective in fat digestion
3.) Bile release
- *Triggered by cholecystokinin (CCK)*, released from the duodenum when fatty foods enter
    - Contraction of duodenum -> cystic duct -> CBD -> release of bile
- Bile salts emulsify fats, breaking them into micelles for pancreatic lipase to act upon

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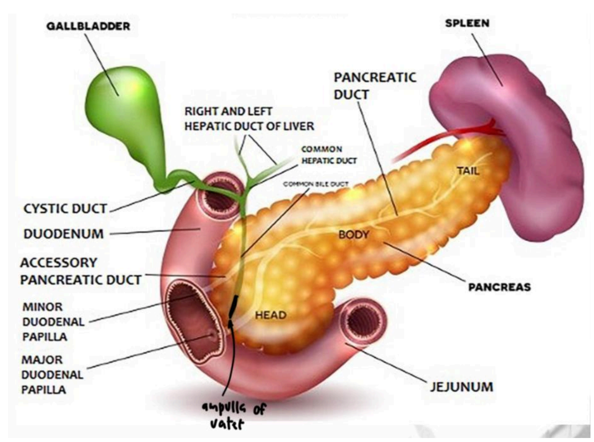
Pancreas



- Vital accessory organ of GI and endocrine system

WEEK 9: ORGAN SYSTEM

- **Retroperitoneal organ** - located behind the stomach, in the upper abdomen
    - Extends horizontally across the posterior abdominal wall from the duodenum (right side) to spleen (left side)
- **Divided into:**
    - ***Head*** - nestled in the curve of the duodenum
    - ***Neck*** - short, between head and body
    - ***Body*** - extends leftward across midline
    - ***Tail*** - touches the hilum of the spleen (graveyard of RBC)

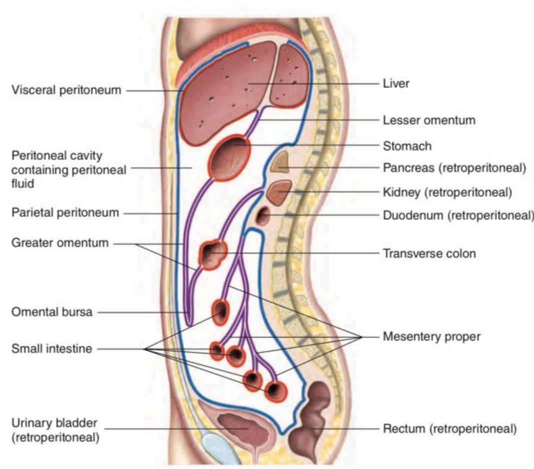
FUNCTIONS OF PANCREAS: Dual function
1.) Exocrine (digestion of nutrients)
- Makes up ~98% of the pancreas
- Acinar cells secrete digestive enzymes
    - Collected into ducts
    - Delivered to duodenum via pancreatic duct (duct of Wirsung)
- Enzymes include:
    - Proteases (carbopeptidase, chymotrypsin, trypsin) - break down proteins
    - Amylase - break down carbohydrates
    - Lipase - break down fats
    - Nuclease - digest DNAs and RNAs
- ***Secretes bicarbonate-rich fluid from ductal cells*** -> neutralizes acidic chyme from stomach
- Secretion is regulated by intestinal hormones:
    - Secretin - stimulates bicarbonate release
    - Cholecystokinin - stimulates enzyme release

2.) Endocrine (regulation of blood glucose and metabolism)
- Carried out by Islets of Langerhans (for blood glucose homeostasis), clusters of hormone-producing cells scattered throughout the pancreas
- Only ~2% of pancreas, but critical in metabolism
- ***Major cell types:*** GABIDS
    - Alpha cells - glucagon, raises blood glucose
    - Beta cells - insulin, lowers blood glucose
    - Delta cells - somatostatin, inhibits both insulin and glucagon release
    - PP cells (F cells) - pancreatic polypeptide, regulates pancreatic secretion

CLINICAL RELEVANCE
1.) Diabetes mellitus
- Type I - autoimmune destruction of beta cells -> increase blood glucose
- Type II - insulin resistance and relative deficiency

2.) Pancreatitis
- Inflammation of pancreas (commonly due to gallstones or chronic alcohol use)
- Enzymes are activated within pancreas, leading to self-digestion

PERITONEUM
- **Visceral peritoneum (serosa)**
    - Serous membrane that covers the organs
- **Parietal peritoneum**
    - Lines the wall of the abdominal cavity
- **Mesenteries**
    - A connective tissue sheet that holds many organs of the abdominal cavity
    - Consist of 2 layers of serous membranes with a thin layer of loose connective tissue between:
- **Retroperitoneal** - other abdominal organs lie against the abdominal wall, *have no mesenteries*
    - Include the duodenum, pancreas, ascending colon, descending colon, kidneys, adrenal glands, and urinary bladder



WEEK 9: ORGAN SYSTEM

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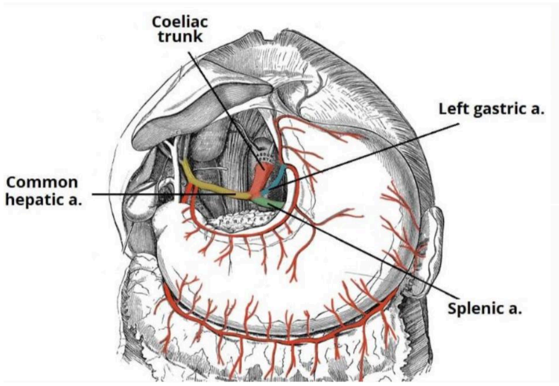
BLOOD SUPPLY OF THE GI SYSTEM

Arterial Supply (from the Abdominal Aorta)

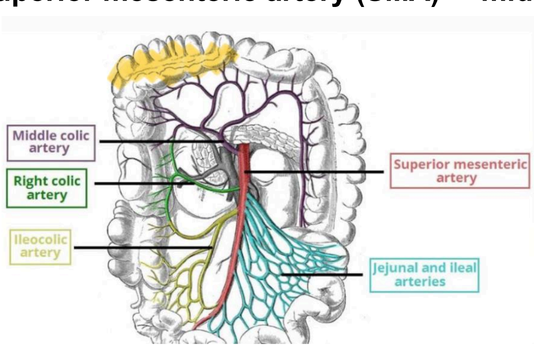
- Embryologically, GI tract is divided into 3 regions:

1.) Celiac trunk - “foregut”



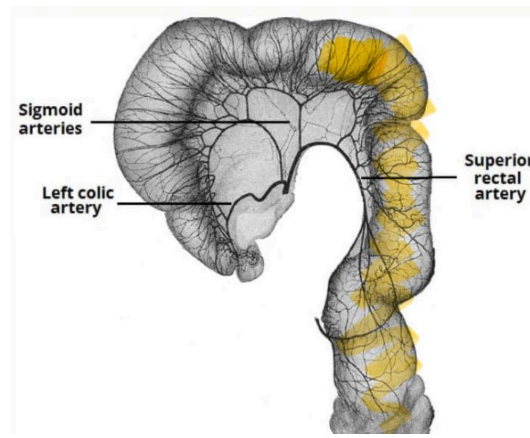
- <u>Location:</u> between T12 (middle suprarenal)
- <u>Supplies:</u> **s**tomach, **s**pleen, **l**iver, **e**sophagus (abdominal part), **p**ancreas, and **p**roximal duodenum (near the junction of the stomach), **g**allbladder - **SSLEPPG**
- <u>Main branches:</u>
  - ***Left gastric artery*** - supplies stomach, lower esophagus
  - ***Splenic artery*** - supplies left stomach, spleen, pancreas
  - ***Common hepatic artery*** - supplies stomach, liver, duodenum, gallbladder
- <u>If damaged:</u> dysfunction of bolus

2.) Superior mesenteric artery (SMA) - “midgut”



- Location: L1
- <u>Supplies:</u> from distal duodenum —> proximal ⅔ of transverse colon (**j**ejunum, **i**leum, **c**ecum, **a**ppendix, **a**scending colon) - **JICAA**
- <u>Main branches:</u>
  - ***Intestinal arteries*** - jejunum, ileum
  - ***Ileocecal artery*** - supplies cecum, appendix, terminal ileum; prevents bleeding during surgery
  - ***Right colic artery*** - supplies ascending colon
  - ***Middle colic artery*** - supplies proximal ⅔ of transverse colon

3.) Inferior mesenteric artery (IMA) - “hindgut”



- <u>Location:</u> L3
- <u>Supplies:</u> from distal ⅓ of transverse colon —> rectum and upper anal canal
- <u>Main branches:</u>
  - ***Left colic artery*** - supplies descending colon
  - ***Sigmoid artery*** - supplies sigmoid colon
  - ***Superior rectal artery*** - supplies rectum

Venous Drainage (Portal Venous System)

- Portal circulation in GI tract

1.) Portal vein formation:

- Superior mesenteric vein + splenic vein (often receives the inferior mesenteric vein)

2.) Drainage pathway:

GI organs -> veins -> portal vein -> liver sinusoids (for nutrient processing & detoxification) -> hepatic veins -> IVC -> RA

WEEK 9: ORGAN SYSTEM

- This system ensures that absorbed nutrients (and toxins) from the intestine first pass through the liver before reaching the systemic circulation

CLINICAL RELEVANCE:
1:) Portal hypertension
- Caput medusae

3.) Ascites
4.) Spleenomegaly
5.) Hepatomegaly

INNERVATION OF THE GI SYSTEM

Enteric Nervous System (ENS) - "Brain of the Gut"

- Large network of neurons (~100 million, almost as many as the SC) located in the walls of the GI tract
- It can function independently of the CNS, though it is modulated by the ANS

MAJOR PLEXUSES

1.) Myenteric plexus (Auerbach's) - "OurMyer"
- Located between circular and longitudinal muscle layers of the muscularis externa
- Controls motility - peristalsis, segmentation, and muscle tone

2.) Submucosal plexus (Meissner's)
- Located in the submucosa
- Controls secretion and blood flow in the mucosa

Autonomic Nervous System (ANS) — Extrinsic Control

1.) Parasympathetic innervation
- ***CN 10***: supplies foregut and midgut up to the transverse colon
- ***Pelvic splanchnic nerves (S2-S4)***: supply hindgut (descending colon, sigmoid colon, rectum)

FUNCTIONS:
1.) Stimulates motility (peristalsis, segmentation)
2.) Increases secretions (saliva, gastric juice, pancreatic enzymes, intestinal secretions)
3.) Relaxes sphincters to promote passage of food

2.) Sympathetic innervation
- Originates from thoracolumbar SC (T5-L2)
- Fibers synapse in prevertebral ganglia (celiac, superior and inferior mesenteric ganglia)

FUNCTIONS:
1.) Inhibits digestion - reduces peristalsis, secretion, and blood flow in the GI tract
2.) Contracts sphincters (e.g., LES, anal)
3.) Diverts blood to skeletal muscles during stress

GI Reflexes — *GGROE*

1.) Gastrocolic reflex
- Distension of the stomach after a meal stimulates increased motility of the colon
- Helps create mass movements, often leading to defecation urge after eating

2.) Gastroileal reflex
- Increased stomach activity (e.g., presence of food) enhances peristalsis in the ileum and relaxes the ileocecal valve
- Allows the chyme to move into large intestine

3.) Rectosphincteric reflex
- Rectal distension causes relaxation of the internal anal sphincter
- Initiates the urge to defecate (voluntary by external sphincter follows)

4.) Other reflexes
- ***Peristaltic reflex:*** stretching of intestinal wall -> proximal contraction + distal relaxation
- ***Vagovagal reflex:*** stretch of stomach triggers vagal afferents -> CNS -> vagal efferents -> enhances gastric secretions & motility

5.) Enterogastric reflex

WEEK 9: ORGAN SYSTEM

- Distension and irritation in the small intestine send inhibitory signals to the stomach
- Slows gastric emptying when the duodenum is overloaded with chyme or too acidic
- ***Mediated by:*** ENS + sympathetic nerves + vagus inhibition

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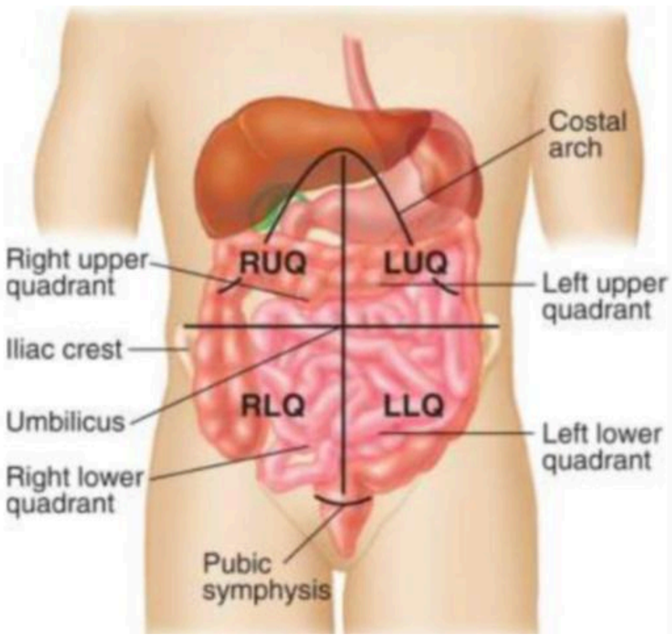
Abdominal Quadrant



Regions of the Abdomen